

RANDOM KNOTS VIA STIEFEL MANIFOLDS

ALEXANDER KOLPAKOV AND IGOR RIVIN

ABSTRACT. A fixed simplex, randomly projected into three dimensions and joined in Hamiltonian order, produces a rich and unusually tractable model of random stick knots. We prove that Gaussian projections and Haar-random Stiefel projections have exactly the same knot-type law, despite having different metric shapes, and that at every stick budget the model gives positive probability to precisely the knot types realizable with that many sticks. Its linear-algebraic structure yields an exact marginal distance law, an exact mean planar-crossing count, and crossing concentration, while in the first nontrivial six-stick case the complete tetrahedral sign pattern gives an exact unknot-versus-handed-trefoil classifier for every generic sample. The result is a direct bridge from random projections and finite sign geometry to the topology of random knots.

1. INTRODUCTION AND SETUP

Assume $N \geq 4$. Let $e_1, \dots, e_N$ be the standard basis of $\mathbb{R}^N$, and set

$$H := \mathbf{1}^\perp \subset \mathbb{R}^N, \quad \Pi_H := I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^T.$$

For a matrix $M \in \mathbb{R}^{3 \times N}$, write $x_i := Me_i$. The Hamiltonian cycle

$$\mathcal{C} = (e_1, e_2, \dots, e_N, e_1)$$

becomes the closed polygon

$$\Gamma(M; \mathcal{C}) := [x_1, x_2] \cup [x_2, x_3] \cup \dots \cup [x_N, x_1] \subset \mathbb{R}^3.$$

When this polygon is embedded, it is a polygonal, or stick, knot with N sticks. Throughout, ambient isotopies start at the identity and hence preserve the orientation of $\mathbb{R}^3$; a chiral knot and its mirror are distinct knot types unless we explicitly merge mirror labels. Knot types themselves are unoriented embedded subsets: reversing the cyclic traversal changes no set $\Gamma(M; \mathcal{C})$ and hence does not change its knot type.

All edge vectors $e_i - e_{i+1}$ lie in H . Thus replacing M by $M\Pi_H$ only subtracts the centroid from the vertices and translates the polygon. This is the basic reason that the centered row space, rather than the absolute position of the simplex vertices, controls the topological part of the construction.

We use two closely related models.

- **Raw Gaussian model:** $P \in \mathbb{R}^{3 \times N}$ has independent entries distributed as $\mathcal{N}(0, 1/3)$, and the polygon is $\Gamma(P; \mathcal{C})$.
- **Haar row-orthonormal model:** Q is Haar distributed on

$$\text{St}_3(H) := \{Q \in \mathbb{R}^{3 \times N} : QQ^T = I_3, Q\mathbf{1} = 0\},$$

and the polygon is $\Gamma(Q; \mathcal{C})$.

The condition $Q\mathbf{1} = 0$ is the centering condition: if q_i is the i -th column of Q , then $\sum_i q_i = 0$. Thus it is a linear dependence among the columns, not among the three rows; $QQ^T = I_3$ says the rows are orthonormal, so $Q|_H$ has rank three. The first model is the direct Gaussian projection of the simplex. The second retains only the random three-dimensional shadow of the centered simplex and removes the singular-value deformation of the Gaussian map.

2. RELATION TO RANDOM-KNOT GENERATION

Many computational random-knot ensembles begin in $\mathbb{R}^3$: one samples lattice self-avoiding polygons, equilateral or Gaussian polygons, confined polygons, or random-jump polygons, and then studies the resulting knot statistics [19, 8, 6, 3, 9]. Other ensembles are diagrammatic, such as Petaluma knots, grid diagrams, and random planar diagrams [11, 12, 13, 4, 5]. The projected-simplex ensemble differs from both families in a specific way. The randomness first chooses a three-dimensional linear shadow of a fixed high-dimensional simplex, and only then reads off a polygon by a Hamiltonian ordering of the vertices. In this respect the closest geometric comparison is Westenberger's random-projection model, where a fixed curve in high-dimensional Euclidean space is projected to random 3-planes [20]. The computations below estimate Stiefel sector probabilities by sampling, not by exact chamber-volume formulas.

Three features are relevant here. First, the topological law has an exact Haar formulation on $\text{St}_3(H)$. Second, the Hamiltonian cycle is distributionally irrelevant: choosing another cyclic ordering only relabels exchangeable simplex vertices. Third, metric effects are separated from topology: raw Gaussian projections add singular-value shear, while the Haar row-orthonormal model removes it without changing knot type.

3. GENERIC STICK KNOTS

Proposition 1 (Generic tameness). *If $M \in \mathbb{R}^{3 \times N}$ has an absolutely continuous law, then $\Gamma(M; \mathcal{C})$ is almost surely an embedded tame polygonal knot. More precisely, with probability one:*

$$x_i \neq x_j \ (i \neq j), \quad \text{no three distinct vertices are collinear,}$$

and no two nonconsecutive edges intersect.

Proof. Coincidence of two vertices means $M(e_i - e_j) = 0$. This is the vanishing of a nonzero linear map of the entries of M , hence a proper linear subspace of $\mathbb{R}^{3N}$ and has Lebesgue measure zero.

For distinct i, j, k , collinearity is equivalent to

$$(x_i - x_k) \times (x_j - x_k) = 0.$$

The three coordinates of this cross product are polynomial functions of the entries of M . They are not all identically zero: one may prescribe the three involved columns to be $(0, 0, 0)$, $(1, 0, 0)$, and $(0, 1, 0)$. Hence the collinearity locus is a proper real-algebraic set and is null.

It remains to treat intersections of nonconsecutive edges. Fix such a pair

$$E_a = [x_a, x_{a+1}], \quad E_b = [x_b, x_{b+1}].$$

Let

$$d_1 = x_{a+1} - x_a, \quad d_2 = x_{b+1} - x_b.$$

If the two segments intersect, then their supporting lines are coplanar, so

$$\tau_{ab}(M) := (d_1 \times d_2) \cdot (x_b - x_a) = 0.$$

This is a degree-three polynomial in the entries of M . It is not identically zero: choose the four relevant columns to be

$$x_a = (0, 0, 0), \quad x_{a+1} = (1, 0, 0), \quad x_b = (0, 1, 0), \quad x_{b+1} = (0, 0, 1).$$

Then $d_1 = (1, 0, 0)$, $d_2 = (0, -1, 1)$, and

$$\tau_{ab} = ((1, 0, 0) \times (0, -1, 1)) \cdot (0, 1, 0) = -1.$$

Thus $\{\tau_{ab} = 0\}$ is a proper algebraic hypersurface and is null. Since there are only finitely many vertex triples and edge pairs, the union of all bad events has probability zero. A polygonal embedded closed curve is tame, so the proposition follows. $\square$

4. HAAR ROW FACTORS AND KNOT TYPE

The topological distribution of the Gaussian model is simpler than its metric distribution. The singular values of the centered Gaussian map change lengths and angles, but they do not change knot type.

Proposition 2 (Gaussian polar decomposition). *Let $N \geq 4$, and let $P \in \mathbb{R}^{3 \times N}$ have independent centered Gaussian entries with common variance $\sigma^2 > 0$. Set $\tilde{P} := P\Pi_H$. With probability one, $\tilde{P} : H \rightarrow \mathbb{R}^3$ has rank three, and its polar row factorization*

$$\tilde{P} = AQ, \quad A := (\tilde{P}\tilde{P}^T)^{1/2} \in GL^+(3, \mathbb{R}), \quad Q \in \text{St}_3(H),$$

has Q Haar distributed on $\text{St}_3(H)$, independently of A , and

$$\frac{A^2}{\sigma^2} \sim W_3(N-1, I_3).$$

Here $W_3(\nu, I_3)$ means the law of $\sum_{r=1}^{\nu} z_r z_r^T$ for independent $z_r \sim \mathcal{N}_3(0, I_3)$. The polygons $\Gamma(P; \mathcal{C})$, $\Gamma(\tilde{P}; \mathcal{C})$, and $\Gamma(Q; \mathcal{C})$ have the same ambient isotopy class. Consequently the knot type is independent of every measurable statistic of the positive factor A , including its unordered singular values and condition number.

Proof. Choose an isometry $B : \mathbb{R}^{N-1} \rightarrow H$ and write $G = \tilde{P}B$. Then G is a $3 \times (N-1)$ matrix of independent $\mathcal{N}(0, \sigma^2)$ entries. The standard Gaussian polar decomposition [17] gives

$$G = AU, \quad A = (GG^T)^{1/2}, \quad U = A^{-1}G \in \text{St}_3(\mathbb{R}^{N-1}),$$

where U is Haar, U is independent of A , and $GG^T/\sigma^2 \sim W_3(N-1, I_3)$. One way to see the independence is to disintegrate the Gaussian density, whose value depends only on GG^T , under the right action of $O(N-1)$. Since $Q = UB^T$, the asserted Haar law and independence follow.

Now A is positive definite and $Q = A^{-1}\tilde{P}$ has orthonormal rows in H . The centered vertices satisfy

$$\tilde{P}e_i = A(Qe_i), \quad i = 1, \dots, N.$$

Thus $\Gamma(\tilde{P}; \mathcal{C})$ is the image of $\Gamma(Q; \mathcal{C})$ under the orientation-preserving linear diffeomorphism $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Finally, $\Gamma(P; \mathcal{C})$ is a translate of $\Gamma(\tilde{P}; \mathcal{C})$. Translations and orientation-preserving linear diffeomorphisms preserve ambient isotopy class. $\square$

Remark 1 (Grassmannian shadow and chirality). The row space $\text{row}(Q)$ is Haar distributed on $\text{Gr}(3, H)$. A row space determines the polygon only up to a left action of $O(3)$: orientation-preserving choices give the same chiral knot type, while an orientation-reversing choice gives the mirror. Thus the full chiral law is most naturally a Haar law on $\text{St}_3(H)$. For metric observables and for knot labels that identify mirror images, the law descends to the Grassmannian.

Proposition 3 (Distributional invariance across cycles). *Let $\mathcal{C}$ and $\mathcal{C}'$ be Hamiltonian cycles on $\{1, \dots, N\}$. Under either the raw Gaussian model or the Haar row-orthonormal model, the corresponding random polygons built from $\mathcal{C}$ and $\mathcal{C}'$ have the same distribution.*

Proof. In the Gaussian model the columns are exchangeable, so relabeling the vertices by any permutation does not change their joint distribution. Since any Hamiltonian cycle can be sent to any other by a permutation of labels, the polygon distributions agree.

In the Haar row-orthonormal model, a permutation matrix $\Pi \in O(N)$ sends one cyclic ordering to the other and preserves $H = \mathbf{1}^\perp$. Haar measure on $\text{St}_3(H)$ is invariant under $Q \mapsto Q\Pi$. The same relabeling argument gives the result. $\square$

Proposition 4 (Mirror symmetry). *Under either model, the probabilities of a knot type K and its mirror mK are equal.*

Proof. Fix $R \in O(3)$ with $\det R = -1$. Left multiplication $Q \mapsto RQ$ preserves Haar measure on $\text{St}_3(H)$, and the resulting polygon is the image of the original polygon under an orientation-reversing linear isometry. It therefore carries the mirror knot type. The Gaussian law is likewise invariant under $P \mapsto RP$, or follows from Proposition 2. $\square$

Proposition 5 (Well-defined knot-type law). *Under either model, the random stick knot has an almost surely well-defined ambient isotopy class. The resulting probability measure on the countable set of polygonal knot types is Borel measurable and does not depend on the Hamiltonian cycle.*

Proof. For the Gaussian model, Proposition 1 gives an embedded tame polygon almost surely. Let Q be the polar row factor of the centered Gaussian matrix. By Proposition 2, Q is Haar distributed on $\text{St}_3(H)$, and $\Gamma(Q; \mathcal{C})$ is related to the centered Gaussian polygon by an invertible linear map. Thus the Haar row-orthonormal polygon is embedded almost surely as well.

Let $\mathcal{E} \subset \text{St}_3(H)$ be the set of frames giving embedded polygons. The preceding paragraph shows that $\text{St}_3(H) \setminus \mathcal{E}$ is Haar null. On $\mathcal{E}$, the vertex coordinates depend continuously on the frame. Standard stability of PL embeddings implies that $\mathcal{E}$ is open and that sufficiently nearby polygonal embeddings are ambient isotopic [10]. This formulation also covers an embedded polygon having a deliberately inserted collinear subdivision vertex; no noncollinearity margin is required.

Knot type is therefore locally constant on this open embedded-polygon set. Thus the map from $\mathcal{E}$ to the set of knot types is locally constant and hence Borel measurable. Assigning any fixed knot type on the null set $\text{St}_3(H) \setminus \mathcal{E}$ gives a measurable map on all of $\text{St}_3(H)$. Proposition 2 transfers this law to the Gaussian model, and Proposition 3 gives independence of the Hamiltonian cycle. $\square$

5. KNOT-TYPE SECTORS AND STIEFEL PROBABILITIES

Let $\Delta_N \subset \text{St}_3(H)$ be the discriminant of frames for which $\Gamma(Q; \mathcal{C})$ is not an embedded polygon. The preceding section shows that Δ_N is Haar null and that the knot-type map

$$\tau_N : \text{St}_3(H) \setminus \Delta_N \longrightarrow \mathcal{K}$$

is locally constant, where $\mathcal{K}$ denotes the countable set of polygonal knot types. Thus every connected component, or chamber, of $\text{St}_3(H) \setminus \Delta_N$ carries a single knot type. The converse need not hold: many chambers can carry the same knot type. The natural one-to-one object is the knot-type sector

$$\mathcal{S}_N(K) := \tau_N^{-1}(K),$$

which is generally a disconnected union of chambers.

Define the Stiefel knot probabilities by

$$p_N(K) := \mu_{\text{Haar}}\{Q \in \text{St}_3(H) : \Gamma(Q; \mathcal{C}) \text{ has knot type } K\}.$$

Equivalently, after assigning any fixed value on the null set Δ_N ,

$$p_N(K) = \mu_{\text{Haar}}(\mathcal{S}_N(K)).$$

Proposition 6 (Support of the Stiefel knot law). *For every knot type K ,*

$$p_N(K) > 0 \iff \text{stick}(K) \leq N.$$

In particular, $p_N(K) = 0$ if $\text{stick}(K) > N$, and

$$\sum_{K \in \mathcal{K}} p_N(K) = 1.$$

The support is finite.

Proof. If $Q \in \text{St}_3(H) \setminus \Delta_N$ represents K , then its columns form an embedded polygon with N sticks. Hence $\text{stick}(K) \leq N$, proving $p_N(K) = 0$ for $\text{stick}(K) > N$.

Conversely, suppose $\text{stick}(K) \leq N$. Choose a polygonal representative of K with at most N sticks and subdivide edges if necessary to obtain exactly N labeled vertices. After an arbitrarily

small ambient isotopy, assume that the centered coordinate rows span a three-dimensional subspace of H . Let $M \in \mathbb{R}^{3 \times N}$ be the centered vertex matrix, so $M\mathbf{1} = 0$ and MM^T is positive definite. Put

$$Q = (MM^T)^{-1/2}M.$$

Then $Q \in \text{St}_3(H)$, and the original centered polygon is the image of the Q -polygon under an orientation-preserving linear diffeomorphism of $\mathbb{R}^3$. Thus Q represents K .

Embedded polygonal knots are stable under sufficiently small perturbations of their vertices. Therefore a nonempty open neighborhood of Q in $\text{St}_3(H)$ is contained in $\mathcal{S}_N(K)$. Since Haar measure has full support on the compact smooth manifold $\text{St}_3(H)$, this open set has positive measure, and $p_N(K) > 0$.

Finally, the map τ_N is Borel measurable by Proposition 5, the discriminant is null, and the countable sectors partition the embedded part of $\text{St}_3(H)$. Hence the probabilities sum to one.

For finiteness, take a generic planar projection of an N -stick polygon. Only non-incident edge pairs can cross, so its crossing number is at most $N(N-3)/2$. There are only finitely many knot diagrams with a bounded number of crossings, and hence only finitely many knot types in the support. $\square$

Remark 2 (Stick-number data). For tabulated exact values and bounds on $\text{stick}(K)$, we use the KnotInfo stick-number table [15]; see also the recent computational updates in [9, 2].

Remark 3 (Topological sectors). The discriminant decomposition is useful for language: knot invariants are constant on each chamber, and $p_N(K)$ is the Haar measure of the union of chambers carrying type K . For $N \leq 5$, stick number leaves only the unknot; at $N = 6$, the only possible nontrivial type is the trefoil.

6. METRIC NORMALIZATIONS

The metric facts below fix the scale of the experiments. The knot-type equivalence between the Gaussian and Haar models does not require a distortion estimate.

Lemma 1 (Raw Gaussian scale). *Let $P \in \mathbb{R}^{3 \times N}$ have independent $\mathcal{N}(0, 1/3)$ entries. For every $i \neq j$,*

$$\mathbb{E}\|P(e_i - e_j)\|^2 = 2.$$

In particular, the raw Gaussian normalization preserves the squared simplex distance $\|e_i - e_j\|^2 = 2$ in expectation.

Proof. The i th and j th columns of P are independent centered Gaussian vectors with covariance $(1/3)I_3$. Their difference is centered Gaussian with covariance $(2/3)I_3$. The trace of this covariance matrix is 2, which is $\mathbb{E}\|P(e_i - e_j)\|^2$. $\square$

Proposition 7 (Haar row-orthonormal scale). *Let $Q \in \text{St}_3(H)$. If $y_i := Qe_i$, then $\sum_i y_i = 0$ and*

$$\frac{1}{\binom{N}{2}} \sum_{1 \leq i < j \leq N} \|y_i - y_j\|^2 = \frac{6}{N-1}.$$

Equivalently, the all-pair RMS distance is exactly

$$\sqrt{\frac{2 \cdot 3}{N-1}}.$$

Proof. The condition $Q\mathbf{1} = 0$ gives $\sum_i y_i = 0$. The standard identity

$$\sum_{i < j} \|y_i - y_j\|^2 = N \sum_i \|y_i\|^2 - \left\| \sum_i y_i \right\|^2$$

therefore gives

$$\sum_{i < j} \|y_i - y_j\|^2 = N \operatorname{tr}(QQ^T) = 3N.$$

Dividing by $\binom{N}{2}$ yields $6/(N-1)$. $\square$

Proposition 8 (Exact marginal distance law). *Let Q be Haar distributed on $\operatorname{St}_3(H)$, let $i \neq j$, and assume $N \geq 5$. Then*

$$Z_{ij} := \frac{\|Q(e_i - e_j)\|^2}{2} \sim \operatorname{Beta}\left(\frac{3}{2}, \frac{N-4}{2}\right).$$

Consequently

$$\mathbb{E}\|Q(e_i - e_j)\|^2 = \frac{6}{N-1}, \quad \operatorname{Var}(\|Q(e_i - e_j)\|^2) = \frac{24(N-4)}{(N-1)^2(N+1)}.$$

For $N = 4$, $Z_{ij} = 1$ deterministically.

Proof. The vector $v = (e_i - e_j)/\sqrt{2}$ is a unit vector in the $(N-1)$ -dimensional space H . The quantity $\|Qv\|^2$ is the squared length of the projection of v onto the Haar random three-plane $\operatorname{row}(Q)$. Rotational invariance identifies its law with

$$\frac{X}{X+Y}, \quad X \sim \chi_3^2, \quad Y \sim \chi_{N-4}^2, \quad X \text{ and } Y \text{ independent},$$

which is the displayed beta distribution. The formulas follow from the standard beta moments. If $N = 4$, the only three-plane in H is H itself. $\square$

Remark 4 (The pair distances are not independent). Proposition 8 is a one-pair marginal statement. Indeed Proposition 7 gives the deterministic dependence

$$\sum_{i < j} \|Q(e_i - e_j)\|^2 = 3N.$$

7. EXACT CROSSINGS OF THE PLANAR SHADOW

Let C_N be the number of proper crossings between non-incident edges in the planar polygon obtained from the first two coordinates of the raw Gaussian model. Equivalently, one may use the first two rows of a Haar row-orthonormal projection: planar polar whitening is an invertible affine map and therefore preserves every segment intersection.

Lemma 2 (Four Gaussian points). *For four independent isotropic Gaussian points in $\mathbb{R}^2$, a prescribed pair of disjoint segments crosses with probability*

$$p_{\times} = \frac{2}{\pi} \arcsin(1/3).$$

Proof. Augment the two coordinate rows of the four points by the all-ones row. The kernel is almost surely a line in $H_4 = \mathbf{1}^\perp$, and that line is uniform because the two centered Gaussian coordinate rows are independent and isotropic in H_4 . Choose a centered Gaussian representative $\lambda = (\lambda_1, \dots, \lambda_4)$ of the line. Its coordinates have pairwise correlation $-1/3$.

The four points are in convex position exactly when the signs of their unique affine dependence split two against two. The trivariate Gaussian orthant formula

$$\Pr(Z_1, Z_2, Z_3 > 0) = \frac{1}{8} + \frac{1}{4\pi} \sum_{a < b} \arcsin(\rho_{ab})$$

shows that any prescribed one-against-three sign cone has probability

$$q = \frac{1}{8} - \frac{3}{4\pi} \arcsin(1/3).$$

There are eight such cones, so the convex-position probability is $1 - 8q = (6/\pi) \arcsin(1/3)$. Conditional on convex position, exchangeability makes each of the three pairings equally likely to be the diagonal pairing. Dividing by three proves the claim. This is the Gaussian instance of Sylvester's four-point problem; see also [14]. $\square$

Theorem 1 (Exact mean and concentration). *For every $N \geq 4$, under either the planar Gaussian or planar Haar model,*

$$\mathbb{E}C_N = \frac{N(N-3)}{\pi} \arcsin(1/3).$$

Moreover, for every $t > 0$,

$$\Pr(|C_N - \mathbb{E}C_N| \geq t) \leq 2 \exp\left(-\frac{t^2}{2N(N-3)^2}\right).$$

Equivalently, with $L_N = N(N-3)/2$, for every $\varepsilon \geq 0$,

$$\Pr(|C_N/L_N - p_\times| \geq \varepsilon) \leq 2e^{-\varepsilon^2 N/8}.$$

Proof. The cyclic N -gon has $L_N = N(N-3)/2$ unordered pairs of non-incident edges. Each indicator has expectation $p_\times$ by Lemma 2, so linearity of expectation gives the mean. Changing one vertex can affect only indicators containing one of its two incident edges, at most $2(N-3)$ indicators. McDiarmid's bounded difference inequality [16] therefore gives

$$2 \exp\left(-\frac{2t^2}{N[2(N-3)]^2}\right),$$

which is the first tail bound; substituting $t = \varepsilon L_N$ gives the second. Finally, the two-row polar decomposition couples the centered Gaussian polygon to the Haar row-space representative by an invertible linear map. The first two rows of a Haar element of $\text{St}_3(H)$ have the Haar law on $\text{St}_2(H)$, proving the Haar statement. $\square$

8. COMPUTATIONAL EXPERIMENTS

The companion **GaussianKnots** repository contains the experiments for the three-dimensional model. It samples the Hamiltonian cycle

$$1 \rightarrow 2 \rightarrow \cdots \rightarrow N \rightarrow 1$$

through projected simplex vertices, identifies the resulting stick knot with **pyknotid** [18], and records both knot type and metric deformation. Two projection models are compared:

- **Haar row-space model:** sample a Haar three-plane in H and choose a row-orthonormal coordinate representative;
- **Raw Gaussian model:** sample $P_{ab} \sim \mathcal{N}(0, 1/3)$, center by $P\Pi_H$, and use the translated-equivalent vertices Pe_i .

The implementation fixes the signs of the QR columns by a deterministic pivot convention. It therefore chooses a representative of a Haar row space rather than a Haar frame on $\text{St}_3(H)$. Every reported metric statistic and every mirror-merged knot label is invariant under the resulting target $O(3)$ change, so the tables estimate the corresponding theoretical quantities. In particular, for a chiral pair these tables estimate the merged probability $p_N(\{K, mK\}) = p_N(K) + p_N(mK)$, not either handed probability separately. For chiral labels one should instead use the positive-diagonal QR convention (or apply an independent Haar element of $O(3)$). By Proposition 2, the two models have the same knot-type probabilities $p_N(K)$. The raw Gaussian run also records the metric shear retained by $P\Pi_H$; this shear can affect diagram complexity and numerical conditioning, so ambiguous classifications are reported separately.

For a fixed N , the natural Monte Carlo estimator of the Stiefel probability is

$$\hat{p}_N(K) = \frac{1}{M} \sum_{r=1}^M \mathbf{1}\{\tau_N(Q_r) = K\},$$

where $Q_1, \dots, Q_M$ are independent Haar samples in $\text{St}_3(H)$. We report the sample size, unknown or ambiguous classifications, and Wilson binomial intervals for the displayed rates. If u of the M numerical classifications are unresolved and x are resolved nontrivial knots, the data themselves only bracket the empirical nontrivial fraction by $[x/M, (x+u)/M]$. The displayed Wilson interval is for the recorded resolved-nontrivial indicator; it is not a correction for classifier uncertainty.

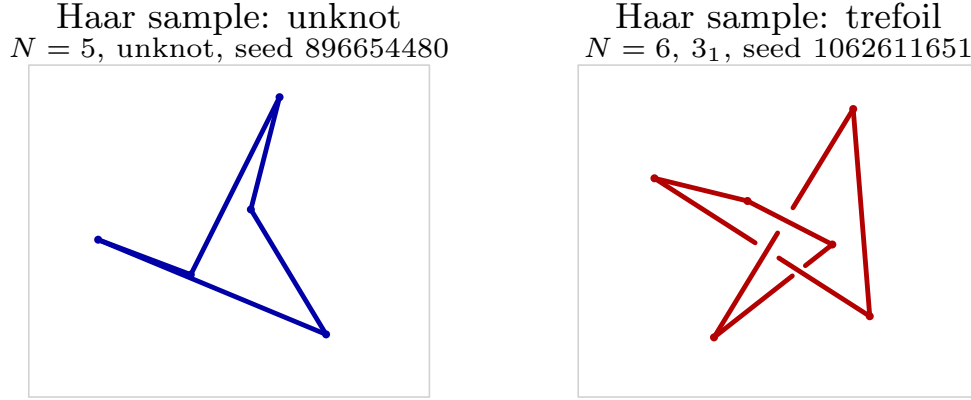


FIGURE 1. Two fixed Haar row-orthonormal samples from the projected-simplex model, drawn as generic planar diagrams with undercrossing gaps. The left sample has $N = 5$ and is unknotted by stick number. The right sample has $N = 6$, the minimum possible stick number for a trefoil, and was identified as 3_1 . The figure is reproducible from the companion script.

For each $N = 5, \dots, 12$ we ran 1000 samples with seed 20260604. The knot catalogue database was installed for `pyknotid`, and the Numba fast backend was available. Table 1 gives the resulting nontrivial-knot rates. The two Monte Carlo runs use matching per-sample seeds and are therefore coupled, not independent. They are still not expected to agree entrywise because their generation maps differ; Proposition 2 gives the same population knot-type law, with finite-sample variation and classification ambiguity visible in the table.

8.1. An exact sign classifier for six sticks. For four distinct vertices define the alternating orientation sign

$$\chi(a, b, c, d) = \text{sgn} \det(v_b - v_a, v_c - v_a, v_d - v_a).$$

At a generic configuration the $15 = \binom{6}{4}$ signs are all nonzero. For five distinct vertices a, b, c, p, q , the signed cofactors

$$\begin{aligned} &(\chi(b, c, p, q), -\chi(a, c, p, q), \chi(a, b, p, q), \\ &\quad -\chi(a, b, c, q), \chi(a, b, c, p)) \end{aligned}$$

are the signs of the coefficients in their unique affine dependence. It follows that the open segment pq pierces the relative interior of the triangle abc exactly when the first three signs agree, the last two agree, and those two common signs are opposite. When this happens, the oriented piercing contribution is $\chi(a, b, c, q)$. This is simply the cofactor identity for the augmented 4×5 vertex matrix, so the test uses only the chirotope.

N	Haar nontrivial	Haar 95% CI	Gaussian nontrivial	Gaussian 95% CI
5	0/1000	0.000–0.004	0/1000	0.000–0.004
6	2/1000	0.001–0.007	5/1000	0.002–0.012
7	13/1000	0.008–0.022	9/1000	0.005–0.017
8	26/1000	0.018–0.038	39/1000	0.029–0.053
9	49/1000	0.037–0.064	38/1000	0.028–0.052
10	66/1000	0.052–0.083	84/1000	0.068–0.103
11	110/1000	0.092–0.131	104/1000	0.087–0.124
12	133/1000	0.113–0.155	128/1000	0.109–0.150

TABLE 1. Empirical resolved-nontrivial counts for the Haar row-space and raw Gaussian models, with nominal Wilson 95% binomial intervals for the displayed indicator. The Haar run had unknown or ambiguous classifications for 1 sample at $N = 10$, 2 samples at $N = 11$, and 1 sample at $N = 12$. The Gaussian run had unknown or ambiguous classifications for 1 sample at each of $N = 9, 10, 11$, and 11 samples at $N = 12$.

Let $I(a, b, c; p, q) \in \{-1, 0, 1\}$ denote that oriented contribution, or zero when there is no piercing, and put

$$\begin{aligned}\Delta_2 &= I(1, 2, 3; 4, 5) + I(1, 2, 3; 5, 6), \\ \Delta_4 &= I(3, 4, 5; 6, 1) + I(3, 4, 5; 1, 2), \\ \Delta_6 &= I(5, 6, 1; 2, 3) + I(5, 6, 1; 3, 4).\end{aligned}$$

Calvo proves for an embedded generic hexagon that all three values are $+1$ precisely for a right trefoil, all are -1 precisely for a left trefoil, and at least one is zero precisely for the unknot [1]. Thus the full 15-sign vector determines the topological knot type; it also determines Calvo’s geometric curl through $\chi(1, 2, 3, 5)$. No connectedness assumption about an order-type realization set is involved.

We applied this exact finite sign rule to the separate 500,000-sample $N = 6$ run, after grouping signatures by Hamiltonian D_6 -symmetry and global sign when only the mirror-merged label was required. The rule found one canonical trefoil signature,

$$+---++-+-++-,$$

and hence

$$\hat{p}_6(3_1) = 1856/500000 = 0.003712, \quad 95\% \text{ CI} = 0.003547\text{--}0.003884.$$

The classifier also agreed on every one of the saved 1000 Haar and 1000 Gaussian $N = 6$ samples with the separately computed `pyknotid` label. The rule is an exact classifier of a generic sign vector; the recorded signs and the Monte Carlo frequency are, of course, numerical. The Wilson intervals for the direct Haar and raw Gaussian counts in Table 1, 2/1000 and 5/1000, are much wider and both overlap 0.003712. The direct runs are therefore consistent with the larger sign-classified run but too small to sharpen the $N = 6$ estimate. In particular, Calvo’s theorem upgrades the bucket labeling, not the Monte Carlo estimate into an exact chamber-volume formula.

These data are not directly comparable to random-polygon or random-diagram ensembles. Equilateral, Gaussian, confined, and quaternionic polygon models sample curves directly in $\mathbb{R}^3$; grid and planar-diagram models start from planar combinatorics. The Stiefel model samples normalized linear shadows of a fixed simplex, and its planar diagrams inherit crossing signs from a height coordinate. Thus the induced shadow and crossing-sign laws differ from the diagrammatic models cited above.

The support theorem gives a deterministic benchmark for finite sampling: every knot type with $\text{stick}(K) \leq N$ has $p_N(K) > 0$, while failure to observe it at sample size M only says that its sector was not hit.

N	0_1	3_1	4_1	5_1	5_2	6_1	6_2	6_3	other	unknown
5	1000	0	0	0	0	0	0	0	0	0
6	998	2	0	0	0	0	0	0	0	0
7	987	13	0	0	0	0	0	0	0	0
8	974	23	3	0	0	0	0	0	0	0
9	951	44	3	0	2	0	0	0	0	0
10	933	57	5	2	1	0	0	1	0	1
11	888	81	12	6	8	1	1	1	0	2
12	866	101	18	5	4	0	0	1	4	1

TABLE 2. Haar row-orthonormal knot-type counts from the same 1000-sample runs. The “other” column records identified nontrivial labels outside the displayed low-crossing types; the “unknown” column records ambiguous catalogue candidate lists or unresolved classifications.

The runs also recorded distance deformation. For all $\binom{N}{2}$ vertex pairs, let

$$D_{\text{pair}} := \frac{\max_{i < j} \|x_i - x_j\|}{\min_{i < j} \|x_i - x_j\|}.$$

The original simplex distance is $\sqrt{2}$. By Lemma 1, the raw Gaussian normalization has expected squared pair distance 2. By Proposition 7, the row-orthonormal Haar model has all-pair RMS distance $\sqrt{2 \cdot 3/(N-1)}$, so it requires the global rescaling factor $\sqrt{(N-1)/3}$ for metric comparison. Table 3 reports mean all-pair max/min distortion, mean RMS distance divided by $\sqrt{2}$, and the mean scale-free all-pair minimum and maximum after per-sample RMS normalization.

N	Haar			Gaussian		
	D_{pair}	$\text{RMS}/\sqrt{2}$	norm. range	D_{pair}	$\text{RMS}/\sqrt{2}$	norm. range
5	2.448	0.866	0.531–1.151	4.335	0.982	0.411–1.492
6	3.467	0.775	0.422–1.267	5.477	0.990	0.342–1.578
7	4.524	0.707	0.362–1.360	6.270	0.988	0.311–1.635
8	5.501	0.655	0.319–1.436	7.244	0.988	0.282–1.693
9	6.384	0.612	0.286–1.500	8.464	0.978	0.256–1.740
10	7.157	0.577	0.260–1.552	9.174	0.988	0.236–1.781
11	8.031	0.548	0.238–1.609	9.958	0.997	0.219–1.813
12	8.632	0.522	0.230–1.648	10.895	0.991	0.206–1.853

TABLE 3. Mean all-pair distance deformation over the same 1000 samples per N . The “norm. range” columns remove one global scale by dividing all projected pair distances in a sample by their RMS. In these runs the normalized ranges show moderate shear and no near-collapse of the vertex set.

9. FORMAL VERIFICATION

The Lean 4/mathlib development at <https://github.com/sashakolpakov/GaussianKnots> defines the Gaussian product measure, proper-crossing statistic, vertex replacement, and six-vertex determinant signature, and verifies the metric identities, raw-centered invariance, $2(N-3)$ sensitivity, and circuit/intersection criterion. From the complete four-point crossing-probability lemma as a typed input, Lean derives the stated mean; from a typed Doob certificate it derives both tail bounds by mathlib’s Azuma–Hoeffding theorem; and it computes all six Calvo records from one 15-determinant signature. The four-point lemma and its orthant argument, construction of the Doob certificate, polar/Wishart transfer, Haar beta law, PL isotopy, and Calvo’s topological classification remain external. The repository contains the declaration-level audit and build instructions.

10. REPRODUCIBILITY

The knot-type and distance-deformation experiments are maintained in the same companion repository. The corresponding Sphinx documentation is published at

<https://sashakolpakov.github.io/GaussianKnots/>.

The main commands used for the tables and the $N = 6$ sign-classified run are

```
python3 scripts/run_knot_experiment.py \
  --projection-model haar \
  --vertices 5,6,7,8,9,10,11,12 \
  --samples 1000 \
  --seed 20260604 \
  --output-dir results/haar_N5-12_1000

python3 scripts/run_knot_experiment.py \
  --projection-model gaussian \
  --vertices 5,6,7,8,9,10,11,12 \
  --samples 1000 \
  --seed 20260604 \
  --output-dir results/gaussian_N5-12_1000

python3 scripts/repro/order_type_grouped_volume.py \
  --vertices 6,7,8 \
  --samples 500000 \
  --seed 20260604 \
  --classify-top-groups 200 \
  --checks-per-group 3 \
  --direct-classify-samples 1000 \
  --output-dir results/order_type_grouped_volume_N6-8_500k
```

The exact-law checks and the Calvo reclassification are reproduced from this note's repository by

```
python3 scripts/repro/validate_exact_projection_laws.py \
  --crossing-samples 10000 \
  --beta-samples 50000 \
  --batch-size 500

GaussianKnots/.venv/bin/python scripts/repro/calvo_n6_exact.py \
  --expected-trefoils 1856 \
  --validate-standard-direct \
  --audit-signs
```

The optional stringent audit replays the complete seeded random-number stream, including the reservoir-sampling draws, and recovers all 155 saved six-vertex sign buckets with no count discrepancy. The closest sampled tetrahedral wall has scale-normalized Plücker margin $3.2516397907 \cdot 10^{-10}$; its affine determinant has magnitude $7.9648583145 \cdot 10^{-10}$, about 7.96 times the sign cutoff. Recomputing the 100 closest cases with 100-digit decimal arithmetic produces no sign mismatch and no canonical-signature mismatch. This is a numerical robustness certificate for the finite sample, not a symbolic evaluation of its Haar probability. Those runs use the performance extra of the `pyknotid` fork, the installed catalogue database, and the Numba fast backend when available.

ACKNOWLEDGMENTS

The authors used OpenAI’s GPT-5.6-sol as a research and coding assistant to help strengthen the results, improve the exposition, and generate and audit the Python and Lean 4 code accompanying this paper. The authors reviewed the resulting mathematics and code and take responsibility for the final content.

REFERENCES

- [1] J. A. Calvo, *Geometric knot spaces and polygonal isotopy*, Journal of Knot Theory and Its Ramifications 10 (2001), no. 2, 245–267. <https://arxiv.org/abs/math/9904037>.
- [2] J. Cantarella, A. Reznitzky, H. Schumacher, and C. Shonkwiler, *New upper bounds for stick numbers*, preprint, arXiv:2508.18263, 2025. <https://www.math.colostate.edu/~clayton/research/papers/stickknots2025.pdf>.
- [3] J. Cantarella, T. Deguchi, and C. Shonkwiler, *Probability theory of random polygons from the quaternionic viewpoint*, Communications on Pure and Applied Mathematics 67 (2014), no. 10, 1658–1699.
- [4] J. Cantarella, H. Chapman, and M. Mastin, *Knot probabilities in random diagrams*, Journal of Physics A: Mathematical and Theoretical 49 (2016), no. 40, 405001.
- [5] H. Chapman, *Asymptotic laws for random knot diagrams*, Journal of Physics A: Mathematical and Theoretical 50 (2017), no. 22, 225001.
- [6] Y. Diao, *The knotting of equilateral polygons in $\mathbb{R}^3$* , Journal of Knot Theory and Its Ramifications 4 (1995), no. 2, 189–196.
- [7] Y. Diao, C. Ernst, and U. Ziegler, *Generating large random knot projections*, in *Physical and Numerical Models in Knot Theory*, Series on Knots and Everything 36, World Scientific, Singapore, 2005, 473–494.
- [8] Y. Diao, N. Pippenger, and D. W. Sumners, *On random knots*, Journal of Knot Theory and Its Ramifications 3 (1994), no. 3, 419–429.
- [9] T. D. Eddy and C. Shonkwiler, *New stick number bounds from random sampling of confined polygons*, Experimental Mathematics 31 (2022), no. 4, 1373–1395.
- [10] R. D. Edwards and R. C. Kirby, *Deformations of spaces of imbeddings*, Annals of Mathematics 93 (1971), 63–88.
- [11] C. Even-Zohar, *Models of random knots*, Journal of Applied and Computational Topology 1 (2017), no. 2, 263–296.
- [12] C. Even-Zohar, J. Hass, N. Linial, and T. Nowik, *Invariants of random knots and links*, Discrete & Computational Geometry 56 (2016), 274–314.
- [13] C. Even-Zohar, J. Hass, N. Linial, and T. Nowik, *The distribution of knots in the Petaluma model*, Algebraic & Geometric Topology 18 (2018), 3647–3667.
- [14] F. Frick, A. Newman, and W. Pegden, *Youden’s demon is Sylvester’s problem*, Mathematika 71 (2025), no. 2, e70015. <https://doi.org/10.1112/mtk.70015>.
- [15] KnotInfo, *Stick number*. https://knotinfo.org/descriptions/polygon_index.html. Accessed June 4, 2026.
- [16] C. McDiarmid, *On the method of bounded differences*, in *Surveys in Combinatorics, 1989*, London Mathematical Society Lecture Note Series 141, Cambridge University Press, 1989, 148–188.
- [17] R. J. Muirhead, *Aspects of Multivariate Statistical Theory*, Wiley, New York, 1982.
- [18] A. J. Taylor and other SPOCK contributors, *pyknotid knot identification toolkit*, <https://github.com/SPOCKknots/pyknotid>, 2017. Accessed May 25, 2026.
- [19] D. W. Sumners and S. G. Whittington, *Knots in self-avoiding walks*, Journal of Physics A: Mathematical and General 21 (1988), no. 7, 1689–1694.
- [20] C. Westenberger, *Knots and links from random projections*, preprint, arXiv:1602.01484, 2019 revision.